Print Server Role Theory – Report (Windows 11 client, HPdeskjet printer)

Server: Server01 (lab.com) | Client: Windows 11 VM | Printer: HPdeskjet

Setup used for the tests

Item	Details
Server	Server01.lab.com, VirtualBox VM, Windows Server build 10.0.26100 (Windows Server 2025), domain lab.com
Client	Windows 11 VM (build 10.0.26200), signed in with the domain accounts lab\student2, lab\student, lab\henry and lab\jack
Printer	HPdeskjet, Generic / Text Only driver, shared as \\server01\HPdeskjet. The printer was kept paused during the tests, so jobs stayed in the queue
Groups	Print group (Student 2, test user), EventManagers (Henry), Volunteers (Jack). Student is in no print group

Task 1: Install the Print Server role on your Windows Server and add a new network printer using the Print Management console.

1. Installed the Print and Document Services role (Print Server) on Server01 through Add Roles and Features.

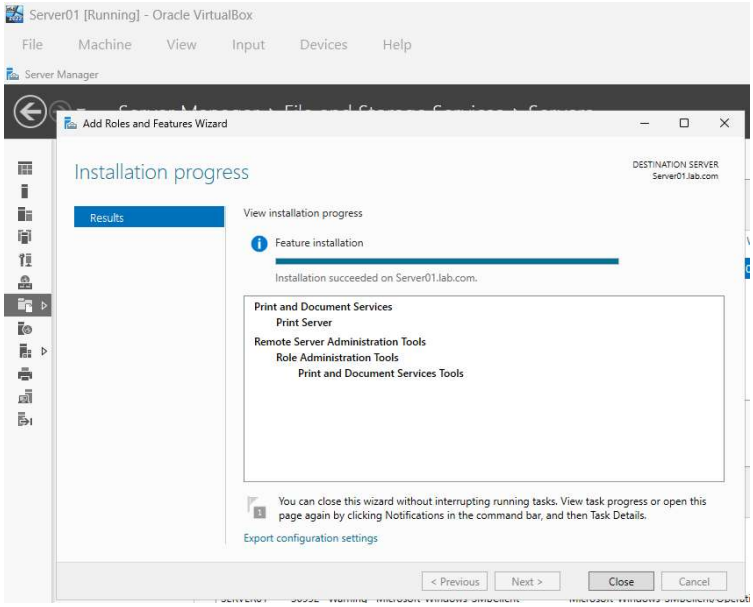


Figure 1: Print Server role installed on Server01.lab.com.

2. Added the printer HPdeskjet with the Generic / Text Only driver and shared it as HPdeskjet, so that clients connect to it at \\server01\HPdeskjet. It is listed in the Print Management console under Printers.

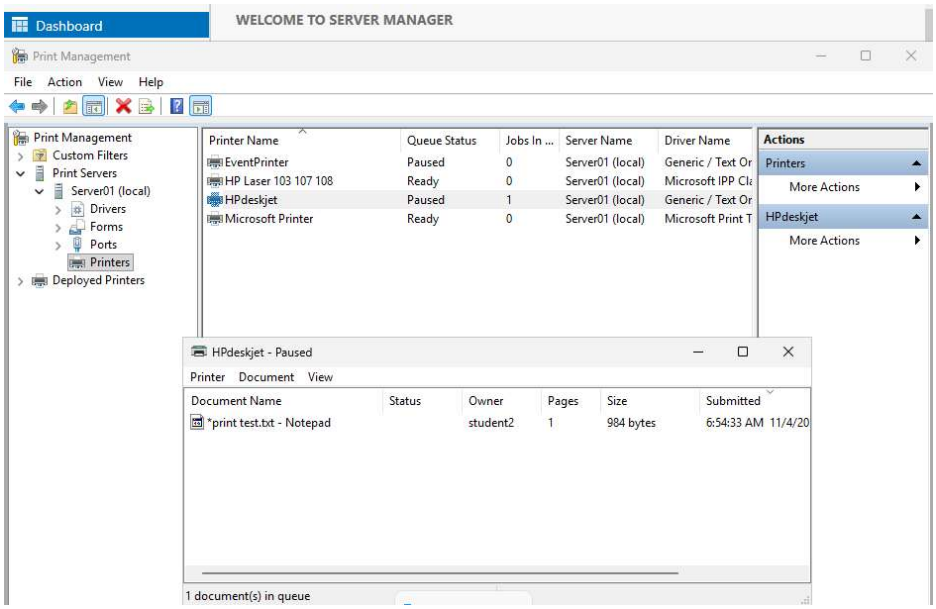


Figure 2: Print Management on Server01 lists HPdeskjet with the Generic / Text Only driver.

Task 2: Configure print permissions for a shared network printer so that only specific users (create two test user accounts) can print, while others are denied access. Hint: Use the Security tab in the printer's Properties to set Allow/Deny permissions for each user.

1. Created the security group Print group in Active Directory and made the two test accounts Student 2 and test user members of it.

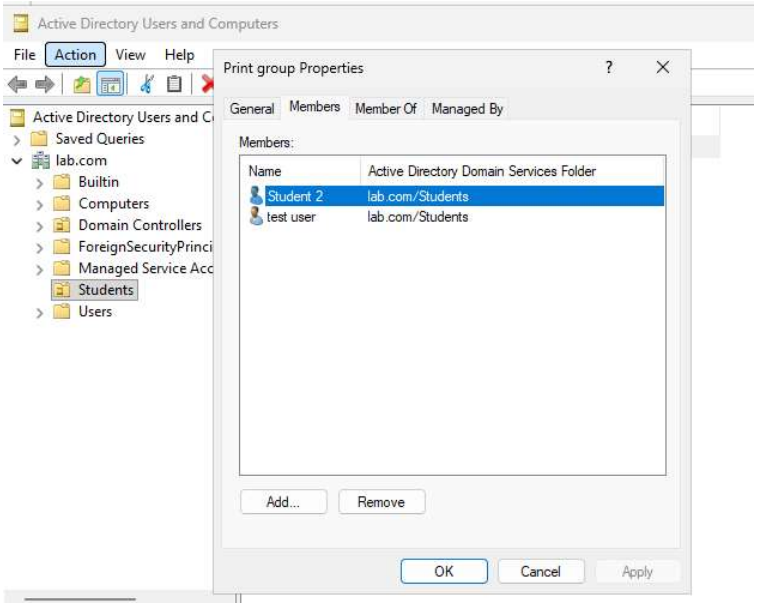


Figure 3: Print group with the members Student 2 and test user.

2. On HPdeskjet's Security tab, removed Everyone, because Everyone has the Print permission by default. Added Print group with Allow Print only. Manage this printer and Manage documents were left unticked.

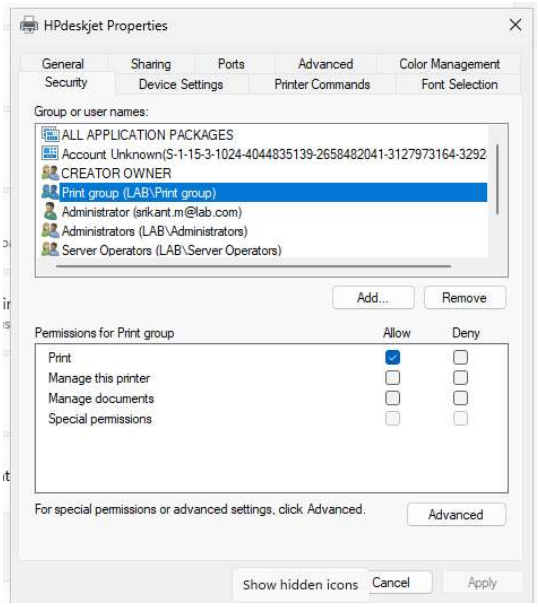


Figure 4: HPdeskjet Security tab: Print group has Allow Print only, and Everyone is not listed.

3. The denied account is Student, who is not in Print group. Windows refuses anything that is not explicitly allowed, so removing Everyone has the same effect as a Deny entry for everyone else. It is also safer than adding a Deny entry for each user, because a Deny overrides any Allow.

Problem seen during this task: when Everyone was removed before Print group had been added, the allowed user student2 could not print from the Windows 11 VM and Windows showed error 0x80070005, which means access denied. This proved that the permission list was being enforced. Adding Print group with Print fixed it.

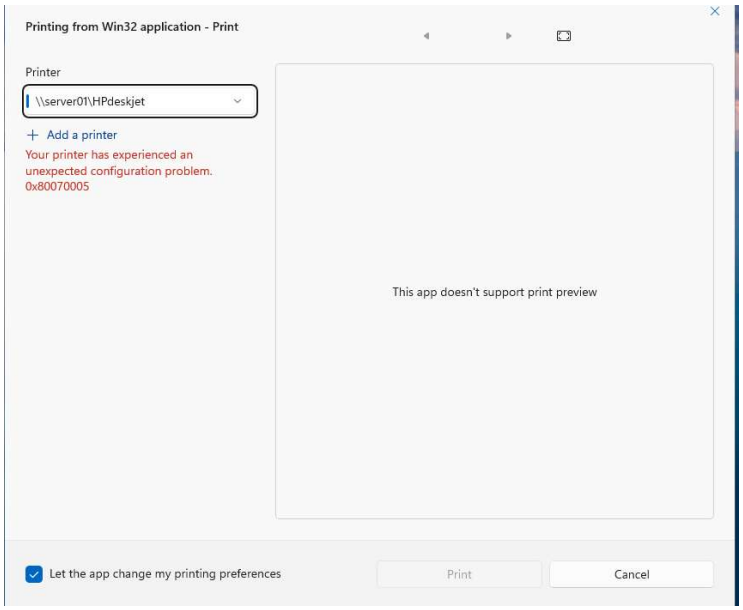


Figure 5: Printing as student2 before Print group was added: error 0x80070005 (access denied).

Task 3: Send a test print job from both an allowed and a denied user account, then use the Print Management console to monitor and manage the print queue, noting what happens for each user.

Both tests were done from the Windows 11 VM by adding the shared printer \\server01\HPdeskjet. The printer was kept paused, so jobs stayed in the queue and could be seen and managed.

Allowed user (lab\student2, a member of Print group):

1. Added \\server01\HPdeskjet by its shared name. The driver installed without any prompt, and the printer was added successfully.

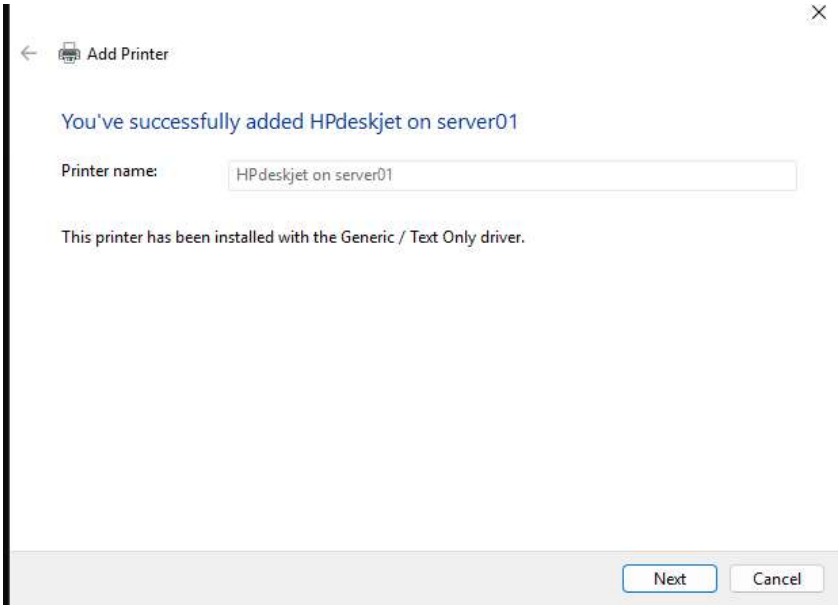


Figure 6: student2 added HPdeskjet on server01 (Generic / Text Only driver).

2. Printed a test page. Print Management on Server01 shows the job in the HPdeskjet queue with the owner student2 and the status Paused.

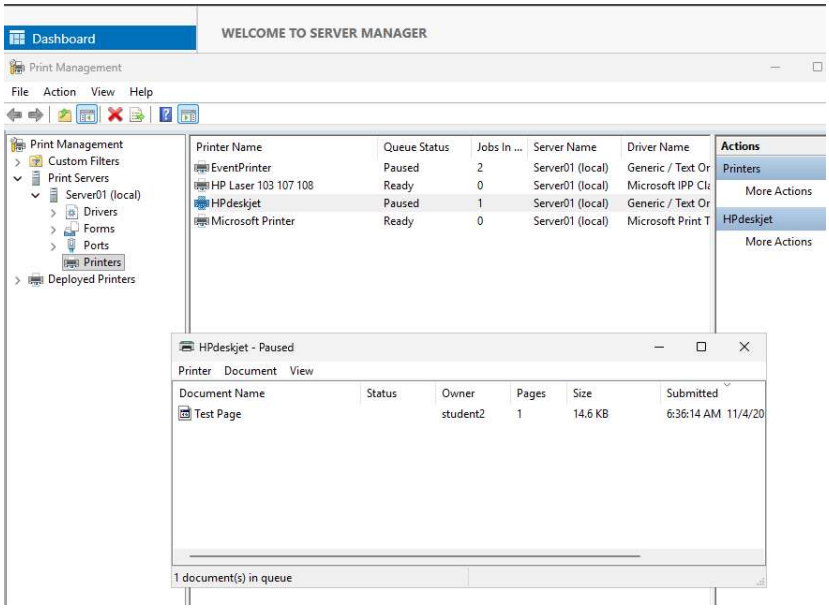


Figure 7: Print Management: the test page from student2 waiting in the HPdeskjet queue.

Denied user (lab\student, not in Print group):

1. Tried to add \\server01\HPdeskjet. Windows refused the connection with the message “The credentials supplied are not sufficient to access this printer”, so the user could not print.

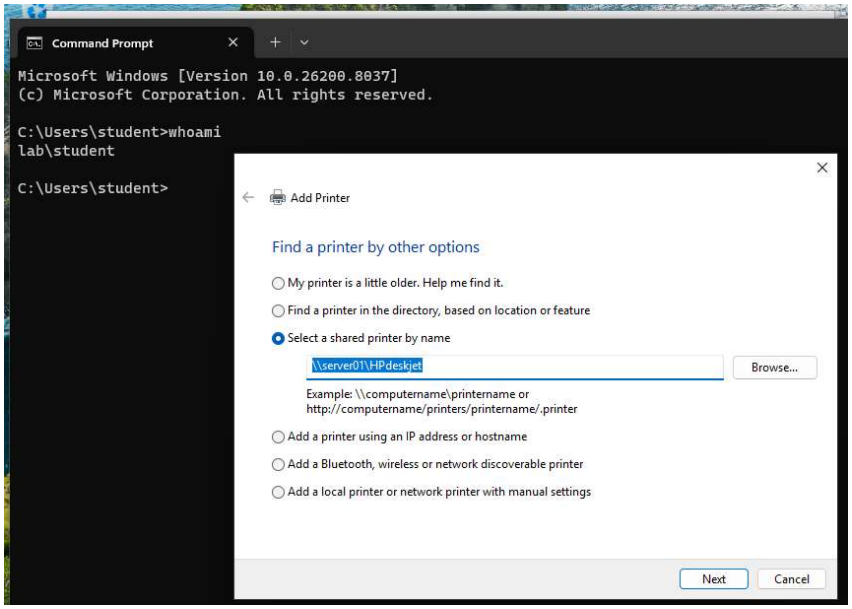


Figure 8: Signed in as lab\student and adding \\server01\HPdeskjet.

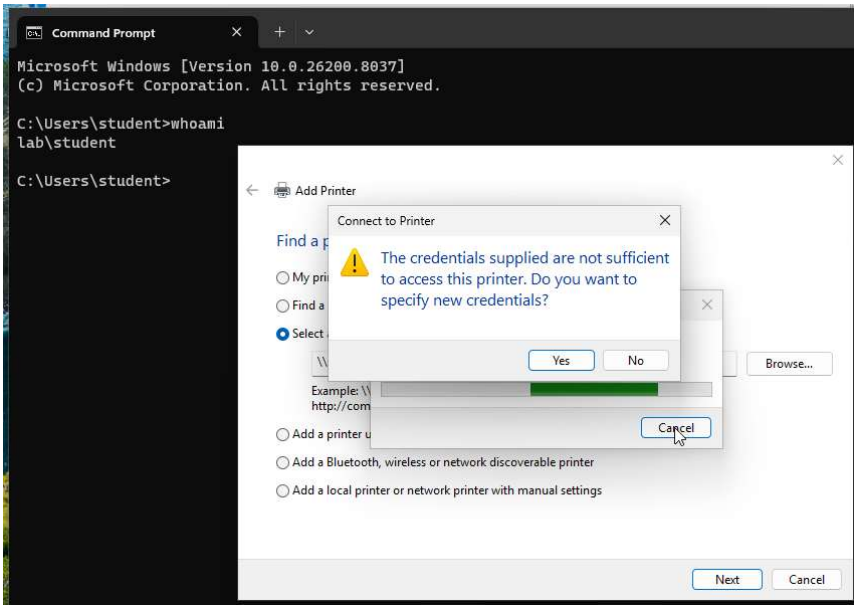


Figure 9: Access to the printer is refused for lab\student.

User	Member of Print group	Result
lab\student2	Yes	Printer added, job reached the server queue
lab\student	No	Access refused, no job sent

Task 4: Change the priority of two print jobs in the print queue so that one user's job prints before another's, regardless of submission order. Hint: Right-click the job in the print queue and adjust the priority value.

With HPdeskjet paused, two documents were submitted. In Print Management, the later job's properties were opened (right-click the job, then Properties, then General) and its priority was moved from 1 to 99.

Job	Submitted	Priority	Position in queue
First job (984 bytes)	6:54:33 AM	1	Second
Second job (1148 bytes)	6:56:54 AM	99	First

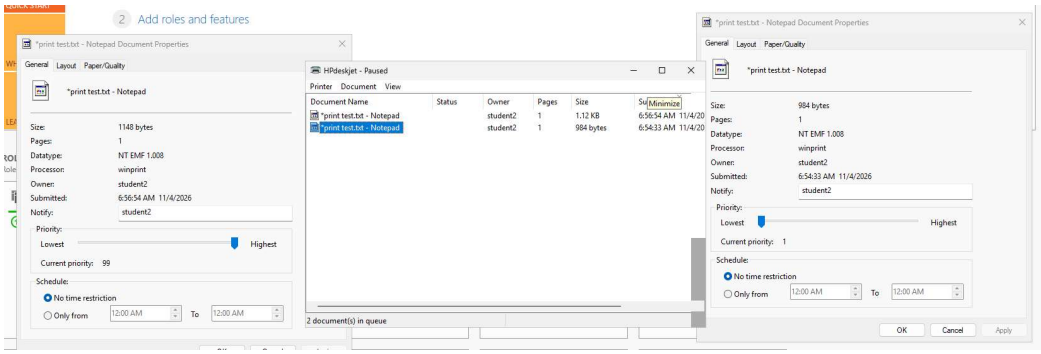


Figure 10: The job submitted later has priority 99 and is listed first. The earlier job keeps priority 1.

A higher priority number is processed first, whatever the submission order, so the second job is printed before the first one once the printer is resumed. Both test jobs were owned by student2. The same setting works between jobs from different users. The printer stayed paused during the test, so the actual printing order was not captured.

Task 5: Simulate a real-world scenario: Imagine your college fest team is printing event posters. Configure the printer so that 'Event Managers' group can manage documents (pause, resume, cancel), while 'Volunteers' can only print, and test these permissions by logging in as both roles.

1. Created two security groups. EventManagers has the member Henry, and the Volunteers group has the member Jack. The Volunteers group was created with the spelling "Volunters", which does not affect how the permissions work.

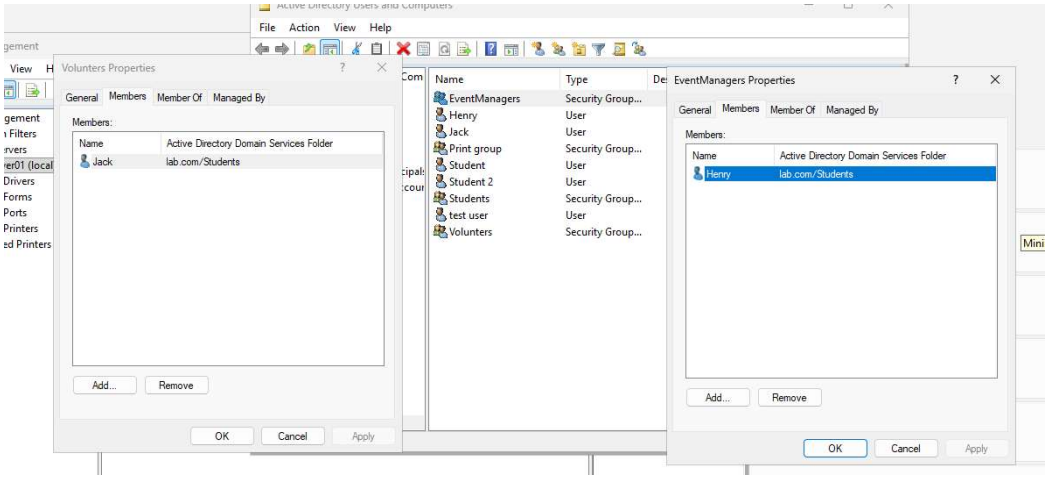


Figure 11: Members of the Volunteers group (Jack) and of EventManagers (Henry).

2. Set the permissions on HPdeskjet's Security tab:

Group	Print	Manage documents	Manage this printer
EventManagers	Allow	Allow	Not allowed
Volunteers	Allow	Not allowed	Not allowed

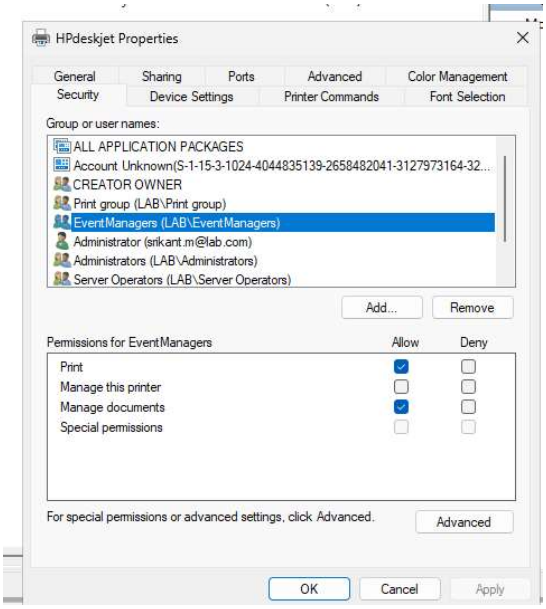


Figure 12: EventManagers: Print and Manage documents allowed.

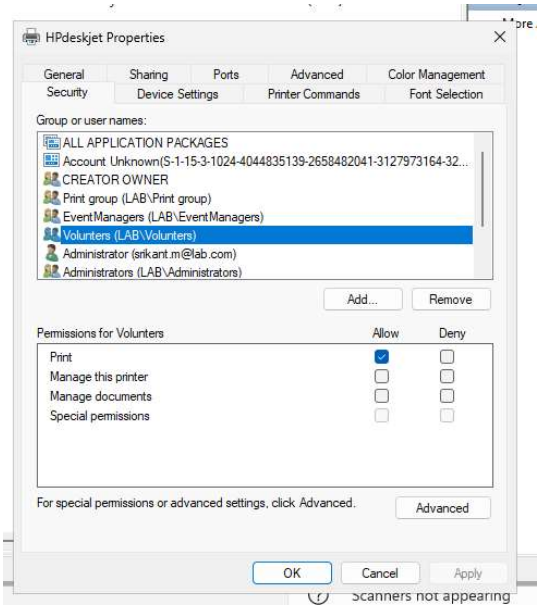


Figure 13: Volunteers group: Print allowed only.

3. Tested as the Event Manager (lab\henry): printed the document Henry Job from Notepad, opened the print queue and used its Pause option on the job. The job's status changed to Paused, and the same status is shown in Print Management on the server.

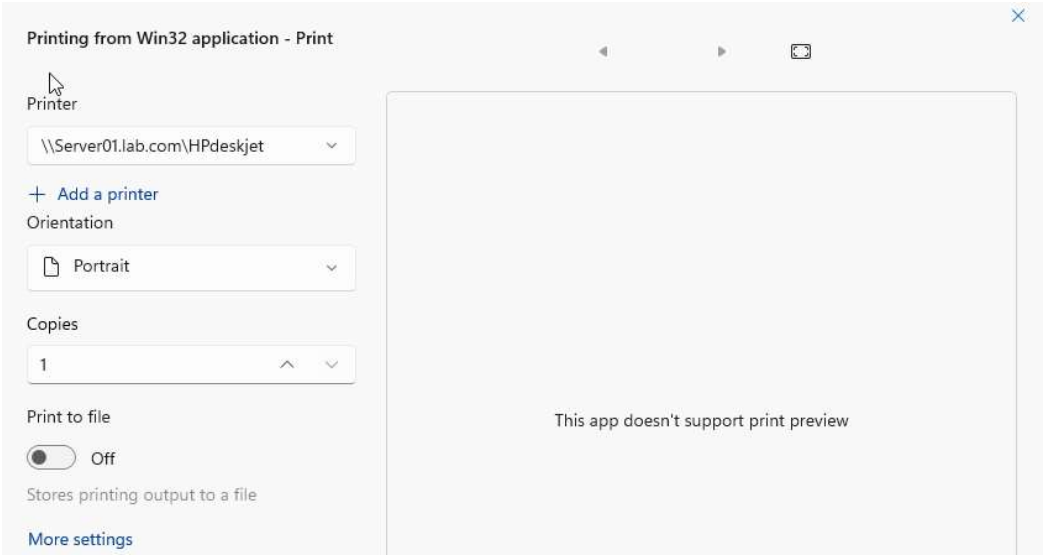


Figure 14: henry printing to \\Server01.lab.com\HPdeskjet.

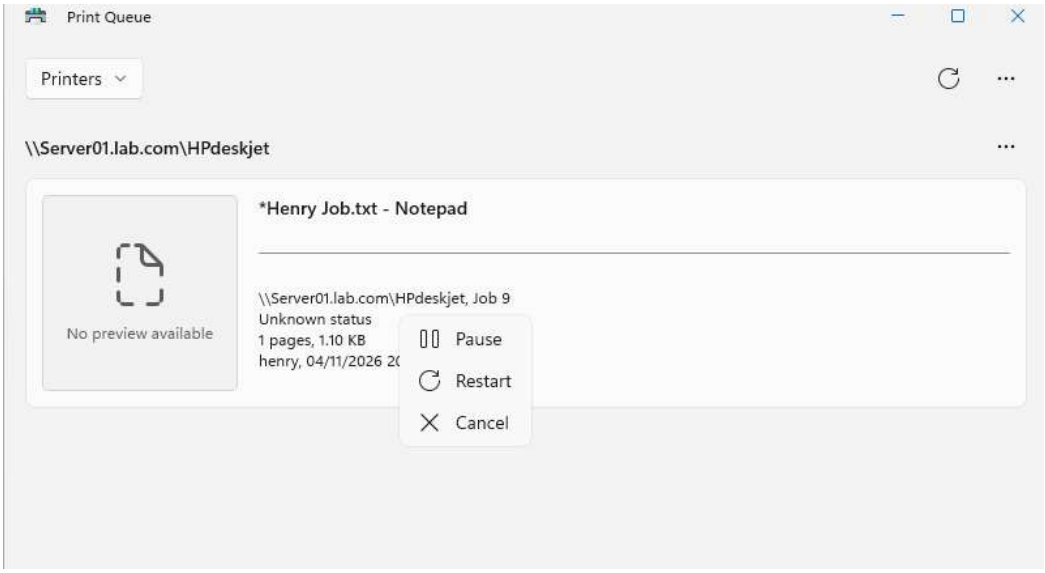


Figure 15: henry's print queue offers Pause, Restart and Cancel for the job.

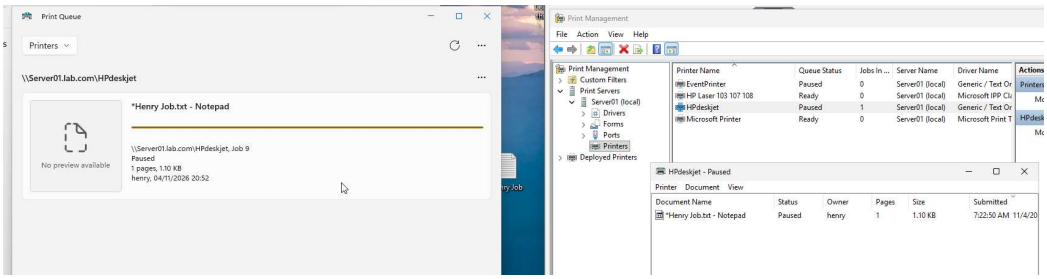


Figure 16: The paused job is shown in the queue (left) and in Print Management on the server (right).

4. Tested as the Volunteer (lab\jack): printed the document Jack Job. Both Jack Job and Henry Job are in the queue.

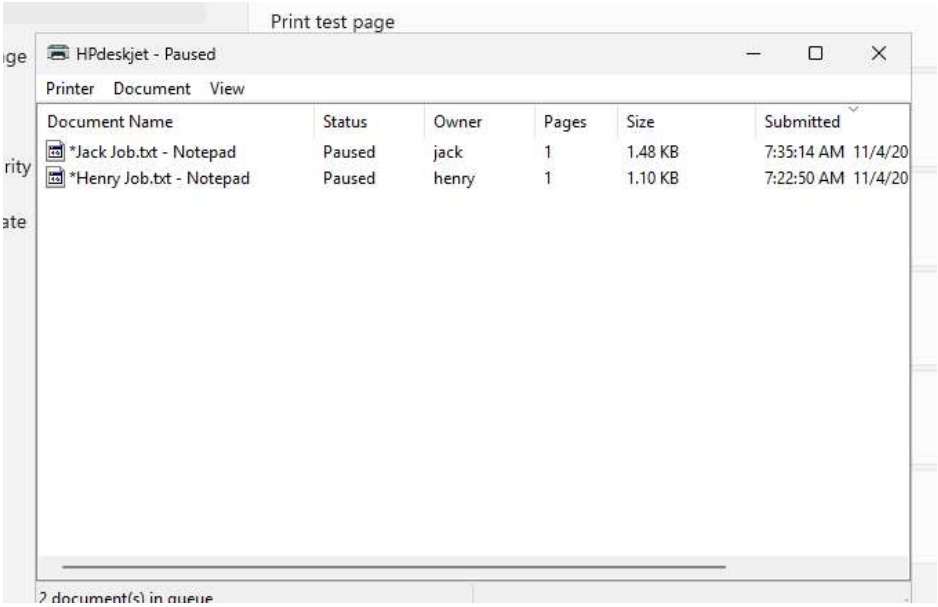


Figure 17: HPdeskjet queue with Jack Job and Henry Job.

Manage documents lets a user pause, resume and cancel documents in the queue, including other users' documents, so it was given only to Event Managers. Volunteers have only Print, so they can submit documents, and as the owner of a job (CREATOR OWNER holds Manage documents) they can manage their own jobs. Neither group has Manage this printer, so neither can pause or change the printer itself.